
ISRAEL VALLEJO MUÑOZ

MARINE BIOLOGIST | ENVIRONMENTAL & SPATIAL DATA ANALYST

• Mexico City ♦ +52 5527718407 ♦ israelvallejo.mz@gmail.com ♦ [LinkedIn](#) ♦ [Portfolio](#)

PROFESSIONAL SUMMARY

Quantitative Marine Ecologist and Environmental Data Analyst specialized in environmental modeling, spatial analysis, and data integration. Experienced in applying R and GIS to analyze climate-driven patterns, fisheries data, and ecological systems, translating complex datasets into actionable insights for environmental decision-making and sustainability applications.

SKILLS

- **Statistical & Ecological Modelling:** R (ggplot2, terra, sf, dismo, ENMeval), CPUE estimation, effort-standardization, regression modelling, time-series analysis, habitat suitability modelling.
- **Geospatial Analysis:** QGIS, Google Earth Engine, spatial data processing, environmental mapping.
- **Environmental & Field Methods:** Marine and coastal field surveys, biodiversity monitoring, laboratory sample analysis, scientific diving (PADI/NADD).
- **Programming & Data Analysis:** R, Excel, Python (basic), RStudio, GitHub.
- **Languages:** Spanish (C2), English (C1), Italian (B2)

WORK HISTORY

Scientific Data Analyst / Fisheries Data Analyst, 01/2026 - Current

Sea Shepherd Global – Remote

- Analyzed operational fisheries datasets including boarding and enforcement records.
- Conducted CPUE estimation and effort-standardized analyses to assess fishing patterns and ecological impacts.
- Evaluated spatial and compliance-related indicators to support conservation and monitoring strategies
- Communicated analytical findings and methodological considerations to international teams.

Environmental Data Analyst, 01/2025 - Current

(Freelancer) – Remote

- Developed climate-driven habitat and spatial analyses using R and GIS.
- Translated complex environmental data into actionable insights to support sustainability and environmental decision-making.
- Conducted data cleaning, preprocessing, and integration of environmental datasets (spatial and temporal).

Intern, 02/2024 - 10/2024

Endocrinology Laboratory, Università Degli Studi D – Naples, Italy

- Prepared and analyzed tissue samples from aquaculture and wild fish species for biochemical and histological assessments.
- Conducted spectrophotometric and chromatographic assays of antioxidant profiles.
- Applied advanced histological techniques for reproductive studies.
- Supported research on the physiological health of marine species.

Teaching Assistant, 04/2024 - 09/2024

Università Degli Studi Di Napoli Federico II – Naples, Italy

- Assisted in laboratory sessions and guided student experiments in marine biology and aquaculture.
- Supported evaluation and review of scientific reports and assignments.

Research Assistant, 07/2018 - 08/2018

Shark Research and Conservation Program – Miami, USA

- Captured and tagged sharks including *Carcharhinus leucas* and *Sphyrna mokarran* (among other species) for ecological and physiological research.
- Collected and processed biopsy, blood, and morphometric data.

Research Assistant, 04/2017 - 01/2018

National Fish Collection – Mexico City, Mexico

- Performed morphometric analysis and skeletal clearing techniques for bony fish and sharks.

Research Assistant, 03/2016 - 01/2017

Instituto de Ciencias del Mar y Limología – Mexico City, Mexico

- Conducted taxonomic identification of demersal fish species from the Mexican Pacific.
- Performed biological dissections and extraction of otoliths for age and growth studies.
- Assisted in laboratory processing, documentation, and organization of ichthyological samples.

EDUCATION

Master of Science: Marine Biology and Aquaculture, 10/2024

Università Degli Studi Di Napoli Federico II - Naples, Italy

- Thesis Statement: Focused on morphological and anatomical responses of thermophilic species under increasing temperature regimes.

Bachelor of Science: Biology, 01/2021

Universidad Nacional Autónoma De México - Mexico City, Mexico

- Thesis Statement: Reproductive Aspects of the Deep-water Shark *Etmopterus schultzi* (Bigelow, Schroeder, Springer, 1953) in the Southern Gulf of Mexico.